NOTES FROM THE FIELD

Telecom data, explained

What six years at the Middle East's biggest mobile service provider taught me about customers, charging systems and ETL

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01 · BUSINESS FIRST

Understand the business before the data

As a PL/SQL developer at the Middle East's biggest mobile service provider, turning raw data into accurate information started with a deep understanding of:

- How customer, account and subscriber relate
- B2C and B2B customers
- International calls and roaming: definition and configuration
- Credit limits and debt collection for postpaid
- Loading and storing Event Detail Records (EDRs)
- Legacy and historical CRM data

02 · THE STARTING POINT

Customer, account, subscriber

Adopting a new business operation system is challenging, and a great chance to learn and grow. It starts with three concepts and the system that owns each one.

Customer

A person who bought at least one SIM, used the service at least once and shared basic information. Can hold several accounts and subscribers, within the regulator’s limits.

Owner: CRM

Account

A wallet that makes usage possible for a SIM. Each SIM has one main account and can have sub-accounts to control usage.

Owner: OCS

Subscriber

A SIM that gives the customer access to services, with one main account and optional sub-accounts. OCS controls account-related usage; CRM steps in for customer-service reasons.

Owner: OCS + CRM

CRM and OCS store data differently

CRM

Less normalized than OCS, which makes it better for reporting UIs.

Online charging system (OCS)

In-memory database (unstructured)

Fast answers for high-demand requests: subscriber status, remaining credit, live usage.

Physical database (structured)

Backs the OCS UI: transactions, payment records and more.

Both UIs call APIs to return information and modify data in the tables.

One prepaid cycle, step by step

B2C serves individuals through three service types: prepaid, postpaid and hybrid. The main difference is how the system treats the main account.

1

Buy a SIM

Pay for the physical SIM only. No deposit.

2

Recharge first

Top up the main account before any pay-as-you-go use. No balance and no free units: emergency services only.

3

Calculate

Remaining balance and free units come from EDRs, in the server’s in-memory DB.

4

Control

OCS is the system owner that controls what the subscriber can use.

Postpaid runs on a credit limit

Available Credit Limit (ACL)

Depending on the provider, a mix of:

+ Refundable and non-refundable deposits
+ Initial gift credit
+ Gift credit for subscriber behavior

– Open bills
– This cycle’s unbilled amount

Buy a SIM: SIM plus a non-refundable deposit

Use, then pay: usage is capped by the ACL; a bill arrives each cycle

AR database: stores deposits, bills and payments

In-memory DB: calculates the remaining ACL at high speed

OCS: the system owner that controls usage

My definition of ETL

“A well-designed, layered system that cleanses, organizes and manages data, aware of each system owner and how data flows through the sources.”



📖 **Don’t miss:** good feedback and documented solutions, so the process is reviewed and updated as the sources change.

It’s a way of thinking

AI’s answer: “turning data into useful insights that help organizations make better decisions.” For me, it’s how I make decisions in daily life, from the simple to the complicated.

SIMPLE

Which gas station?

Petro-Canada, Shell or Esso, and which pump: features, distance, road conditions.

A BIT HARDER

A pair of jeans

The feature list grows, and so does the analysis.

TOO MANY FEATURES?

Two paths

Use tools to analyze more samples, or ask people in the industry who know the hidden features (they probably use tools too).

THANKS FOR READING

Let’s talk data

I’m Mehrad Tavanamehr, a data engineer in Calgary with 6+ years of telecom data experience. I turn raw data from CRM, charging and legacy systems into information people can trust.

SQL · Oracle PL/SQL · ETL / ELT · Data quality · Python · Airflow · Power BI

Open to data engineering roles across Canada. Let’s connect.